

Transformed Skin: Changing own skin texture feeling for VR avatar embodiment

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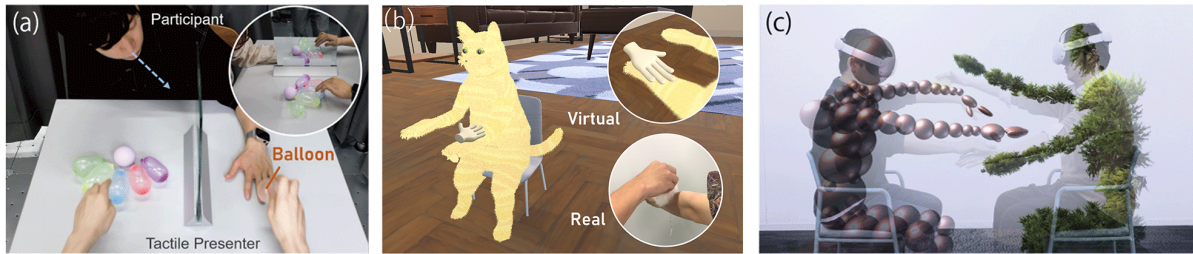


Figure 1: (a) The texture swapping illusion as an embodiment method. (b) users feel their skin texture is changed to the avatars, through repetitive touches (c) interaction between two person with transformed skin.

Abstract

With the development of the metaverse society, avatars are increasingly used for self-expression in games and communication services. However, the visual immersion of avatars often fails to intrinsically transform one's physical body into the avatar, creating a qualitative gap between the two. This gap reduces immersion and the overall quality of the experience. Here we introduce our novel "texture swapping" illusion, where one's skin texture feels as if it has been transformed into textures such as balloon, fur, or leaf. This illusion can be remarkably powerful, with most people feeling its effect even in a mirror condition. In Transformed Skin, we utilise the illusion as a method to enhance embodiment to various avatars. It significantly enhances the sensation of being transformed into the avatar's skin texture and increases the sense of body ownership, leading to new forms of avatar communication and an enriched sense of avatar transformation.

CCS Concepts

• **Human-centered computing** → *Interaction paradigms*.

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SA Emerging Technologies '24, December 03–06, 2024, Tokyo, Japan

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ACM ISBN 979-8-4007-1137-4/24/12

<https://doi.org/10.1145/3681755.3688955>

Keywords

virtual bodies, body illusion, haptic, avatar experience, embodiment

ACM Reference Format:

Sohei Wakisaka, Masatoshi Matsubara, Haruka Onoda, Fu Zening, Lucas Ogasawara, and Kouta Minamizawa. 2024. Transformed Skin: Changing own skin texture feeling for VR avatar embodiment. In *SIGGRAPH Asia 2024 Emerging Technologies (SA Emerging Technologies '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3681755.3688955>

1 Introduction

In VR, we can become various non-human avatars. Avatars become more popular as digital alter egos in gaming and social media. However, while the technology for visual embodiment in avatars advances, our physical selves in the real world rarely align with avatar in terms of skin textures. For instance, individuals adopting non-human avatars like cats or monsters only alter their visual appearance, not their actual physical bodies. This results in a qualitative and structural gap between avatars and physical forms, potentially diminishing immersion and experience quality. To achieve embodiment not only visual but tactile (i.e., avatar skin texture), we propose a method leveraging a newly found illusion named "texture swapping illusion (TSI)," with which we implemented user experiences wherein users feel an avatar's skin texture on their real bodies.

2 PROPOSED METHOD

TSI is a visuo-tactile illusion in which one's skin texture is subjectively replaced with the texture of the objects touching the hand. As shown in Fig.1a, a participant looks at a hand made of balloons through a mirror, and when the tactile presenter gives the experienced participant's hand a tactile presentation with balloons, the participant feels as if his or her hand has been turned into a balloon in terms of skin tactile sensation. The illusion is so powerful that almost all of the dozens of participants who have gone so far reported feeling the illusion. A similar case is the rubber hand illusion [Matthew Botvinick 1998], in which the person feels a sense of bodily possession in response to a rubber hand that is in a different position than the actual hand. However, in TSI mirrors are used to visually match the position of the material hand and the real hand, which is a different experience from the hand position sensory draft of the rubber hand illusion. In addition, the principle of this experience is similar to that of the slime hand illusion, in which we can experience the sensation of the skin freely stretching and contracting [Kodaka et al. 2022]. However, the slime hand illusion is different in that the tactile stimulation is provided by directly touching and physically stretching the participant's hand, whereas in Texture Swapping, the tactile stimulation is provided by a material that haptically matches the texture. It highlights the transformation of material texture and one feels as if one's hands have been transformed into the texture of various textures (Fig.2).



Figure 2: Three examples of avatars and corresponding materials for touching.

2.1 Texture Swapping Illusion (TSI)

2.2 TSI-based Avatar Embodiment

In order to extend the illusion that had previously been created for the hands to the whole body, we created an experience in a VR space. Here, the user wears an HMD and becomes an avatar, and texture swapping is realized by presenting tactile sensations including the whole body, such as the face and feet. As shown in Fig.1b, when the user becomes a cat avatar and receives tactile stimuli in response to movements in VR, the user can feel texture swapping. This allows the user to strongly perceive the sensation of actually becoming fluffy like a cat with the entire body (Fig.3).

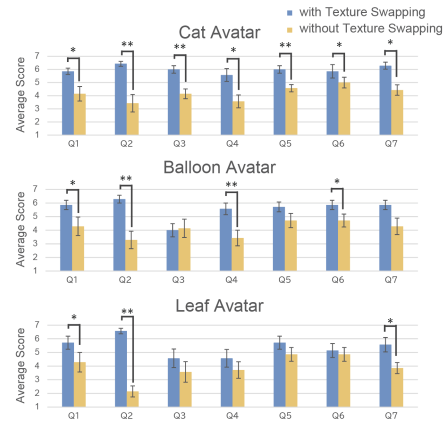


Figure 3: Subjective rating of feelings related to embodiment. Q1. Body ownership, Q2. Tactile texture transformation, Q3. Startle response when stabbed by a virtual knife, Q4. Reality, Q5. Presence, Q6. Immersion, Q7. Body transformation. In "without texture swapping condition," participants were touched by human skin. Wilcoxon Signed-Rank Test. n=7.

3 User Experience

In the demonstration, audience will experience various TSI with analog mirror condition and experience the texture swapping. The user is then invited to experience VR and transform into a character with various sci-fi and inhuman textures, including materials such as foam man and wood man, and animals such as cats, etc. The embodiment method by TSI allows the audience to enjoy a skin transformation that they have never felt before. In the paired version (Fig.1c), two users wearing avatars of different textures can touch each other to mutually feel TSI.

4 Conclusion and Future Work

TSI alters the texture of the own skin of users. This illusion can be generated over the whole-body in immersive VR environments, in various avatars. This not only functions as a powerful trigger to enhance the sense of ownership with the avatar, but may also trigger new behavioral changes. It also has the flexibility to adapt to various textures, as long as realistic, high-quality textures are available. Further detailed verification in the future will expand the possibilities of the avatar experience, leading to new immersive entertainment contents.

Acknowledgments

This work is supported by JST Moonshot R&D Program (Grant number JPMJMS2013)

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